

AI-Based Universal JEE Question Generation System

Multimodal AI Architecture for Automated JEE/NEET Variant Generation

Objective: Develop a scalable AI-native architecture capable of transforming a single expert-authored JEE/NEET question image into unlimited mathematically-valid and visually-consistent question variants using semantic reasoning, symbolic mathematics, and deterministic rendering.

INPUT IMAGE	GPT-4 VISION	JSON ENGINE	QWEN2-VL	FINAL OUTPUT
JEE Question Upload	Semantic Extraction	Constraint Validation	Layout-aware Editing	Generated Variants

1. Problem Statement

Competitive examinations such as JEE Advanced, JEE Mains, and NEET require extensive exposure to conceptually diverse and numerically varied questions. Modern EdTech systems require scalable generation of large high-quality question banks while maintaining educational consistency and mathematical correctness. Traditional manual content authoring requires significant expert effort for question creation, validation, diagram preparation, and formatting. This creates substantial challenges for large-scale personalized learning systems. The proposed architecture addresses this challenge by combining multimodal AI reasoning, symbolic constraint validation, layout-aware image understanding, and deterministic rendering.

Challenge	System Requirement
Large-scale Content Demand	Generate unlimited variants automatically
Mathematical Consistency	Constraint-aware parameter generation
Diagram Preservation	Maintain original visual structure
Semantic Understanding	Extract formulas and relationships accurately
Readable Rendering	Generate clean equations and diagrams

2. System Overview

The architecture operates as a multi-stage AI pipeline. The first stage performs semantic understanding of the original question image using GPT-4 Vision. This stage extracts variables, formulas, mathematical constraints, answer structures, and layout information into a structured JSON representation. The second stage performs symbolic validation using SymPy. Candidate parameters are sampled while ensuring all mathematical and physical constraints remain valid. The third stage uses Qwen2-VL for multimodal layout understanding. The model identifies editable regions and maps newly-generated parameters to the correct visual coordinates. Finally, deterministic rendering engines including Pillow, SVG, and LaTeX-based equation rendering generate high-quality visually-consistent question variants.

INPUT IMAGE	GPT-4 VISION	JSON ENGINE	QWEN2-VL	FINAL OUTPUT
JEE Question Upload	Semantic Extraction	Constraint Validation	Layout-aware Editing	Generated Variants

3. Unified AI Architecture

Layer	Technology	Purpose
Input Layer	Question Image	Original JEE/NEET input
Semantic Layer	GPT-4 Vision	Semantic JSON extraction
Validation Layer	SymPy	Constraint verification
Layout Layer	Qwen2-VL	Region-level edit planning
Rendering Layer	Pillow + SVG + LaTeX	Readable diagram generation
Output Layer	PNG/PDF Export	Final generated variants

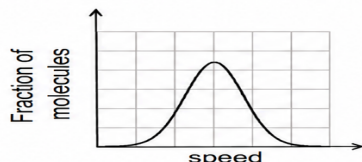
The modular separation between semantic reasoning, symbolic validation, multimodal layout understanding, and deterministic rendering enables scalable generation of mathematically-valid educational content.

4. AI-generated Variant Question Outputs

The following examples demonstrate generated JEE question variants created using semantic JSON extraction, symbolic parameter generation, and layout-aware rendering. The generated outputs preserve educational structure, diagram consistency, and visual readability.

Example 1 — Statistical Mechanics Variant Generation

- Q.1** If the distribution of molecular speeds of a gas is as per the figure shown below, then the ratio of the most probable, the average, and the root mean square speeds, respectively, is

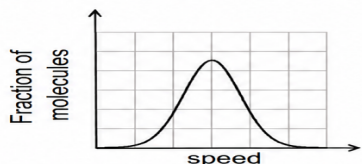


- (A) 1 : 1 : 1
(C) 1 : 1.128 : 1.224

- (B) 1 : 1 : 1.224
(D) 1 : 1.128 : 1

Answer: (C)

- Q.1** If the distribution of molecular speeds of a gas is as per the figure shown below, then the ratio of the most probable, the average, and the root mean square speeds, respectively, is



- (A) 1 : 1 : 1
(C) 1 : 1.150 : 1.250

- (B) 1 : 1 : 1.250
(D) 1 : 1.150 : 1

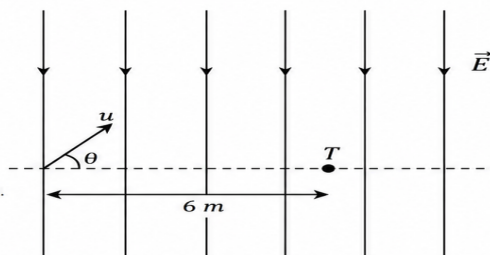
Answer: (C)

Example 2 — Electrodynamics Variant Generation

- 1** A uniform electric field, $\vec{E} = -500\sqrt{3}\hat{y} \text{ NC}^{-1}$ is applied in a region. A positively charged particle having $\frac{q}{m} = 2 \times 10^{10} \text{ Ckg}^{-1}$ is projected with an initial speed $3\sqrt{5} \times 10^6 \text{ ms}^{-1}$. The particle is aimed to hit a target T , which is 6 m away from its entry point into the field as shown in the figure.

Then, the correct statements are

- ☐ A The particle will hit T if projected at an angle 45° from the horizontal.
☐ B The particle will hit T if projected either at an angle 30° or 60° from the horizontal.
☐ C The time taken by the particle to hit T could be $\sqrt{\frac{3}{5}} \mu\text{s}$ as well as $\sqrt{\frac{9}{5}} \mu\text{s}$.
☐ D The time taken by the particle to hit T is $\sqrt{2} \mu\text{s}$.

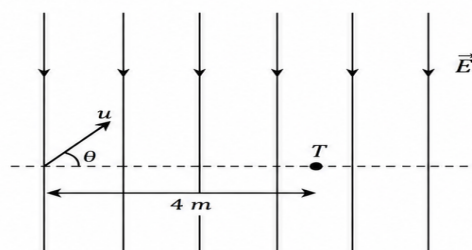


Correct Answer: B, C

- 2** A uniform electric field, $\vec{E} = -300\sqrt{3}\hat{y} \text{ NC}^{-1}$ is applied in a region. A positively charged particle having $\frac{q}{m} = 8 \times 10^9 \text{ Ckg}^{-1}$ is projected with an initial speed $4\sqrt{2} \times 10^6 \text{ ms}^{-1}$. The particle is aimed to hit a target T , which is 4 m away from its entry point into the field as shown in the figure.

Then, the correct statements are

- ☐ A The particle will hit T if projected at an angle 45° from the horizontal.
☐ B The particle will hit T if projected either at an angle 30° or 60° from the horizontal.
☐ C The time taken by the particle to hit T could be $\sqrt{\frac{2}{3}} \mu\text{s}$ as well as $\sqrt{2} \mu\text{s}$.
☐ D Only one trajectory exists for the particle to hit T.



Correct Answer: B, C

5. JSON-guided Variant Generation Workflow

The complete system is driven using a structured semantic JSON pipeline. GPT-4 Vision extracts the semantic representation of the original question. Constraint-aware parameter generation creates mathematically-valid samples. Qwen2-VL performs layout understanding and edit planning, while deterministic rendering engines generate the final variant image.

Input	Transformation	Output
Question Image	GPT-4 Vision Extraction	Semantic JSON
Semantic JSON	Constraint Validation	Validated Parameters
Validated JSON	Qwen2-VL Region Mapping	Edit Instructions
Edit Instructions	Pillow + SVG Rendering	Generated Variant

The architecture enables scalable generation of visually-rich and mathematically-consistent educational content suitable for adaptive learning systems, personalized practice platforms, and AI-assisted EdTech workflows.

6. Conclusion

This architecture demonstrates a complete AI-native framework for automated educational content generation. By combining multimodal semantic reasoning, symbolic mathematics, layout-aware image understanding, and deterministic rendering, the system enables scalable generation of visually-consistent and mathematically-valid JEE/NEET question variants. The modular architecture supports future extensions including adaptive difficulty control, personalized practice generation, multilingual rendering, and large-scale educational dataset generation.